



APPROVED METHOD:
MEASURING LUMINOUS FLUX
WAVEFORMS FOR USE IN TEMPORAL
LIGHT ARTIFACT (TLA) CALCULATIONS
AN AMERICAN NATIONAL STANDARD



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Publication of this report
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared for IES
By the Testing Procedures Committee**



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1.0 Introduction and Scope

1.1 Introduction

Cyclic variations in luminous output from a light source or lighting system are known as temporal light modulation (TLM). TLM is known to affect visual perception, cognition, and brain activity. The variety of TLM patterns of solid-state lighting has brought these effects to the attention of the lighting community. Research seeks to create new metrics that predict the occurrence of visual, cognitive, and neurological effects based on measurements of the physical stimulus, the luminous flux waveform. This Lighting Measurement (LM) standard defines the method to reliably record, accurately measure, and reliably reproduce luminous flux waveforms and data sets from light sources for use in TLA calculations. **Informative Annex A** provides known calculation definitions typically utilized in TLA analysis.

1.2 Scope

This approved method describes the procedures to be followed and precautions to be observed in performing reproducible luminous flux waveform measurements for use in temporal light artifact (TLA) calculations under standard conditions. This document provides a description of the method of measurement, equipment requirements, and formulation of luminous flux waveforms for use in the calculations of temporal light artifacts from light sources. Luminous flux waveforms utilized in TLA calculations are for luminous flux frequencies between 1 Hz and 3,000 Hz under various control conditions. In addition, data reporting formats are described.

Intentionally flashing or signaling lights outside the defined frequency range are not included within the scope of this document.

2.0 Normative References

2.1 CIE TN 006:2016

International Commission on Illumination (CIE). CIE TN 006:2016, Visual Aspects of Time-Modulated Lighting Systems – Definitions and Measurement Models. Vienna: CIE; 2016.

2.2 ANSI/IES LM-9-20

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurement of Fluorescent Lamps. New York: IES; 2020.

For measurements of fluorescent lamps, the laboratory shall meet the requirements stated therein.

2.3 ANSI/IES LM-45-20

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurement of General Service Incandescent Filament Lamps. New York: IES; 2020.

For measurements of general service incandescent filament lamps, the laboratory shall meet the requirements stated therein.

2.4 ANSI/IES LM-51-20

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurements of High Intensity Discharge Lamps. New York: IES; 2020.

For measurements of high intensity discharge lamps, the laboratory shall meet the requirements stated therein.

2.5 ANSI/IES LM-66-20

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurements of Single-Ended Compact Fluorescent Lamps. New York: IES; 2020.

For measurements of single-based fluorescent lamps, the laboratory shall meet the requirements stated therein.

2.6 ANSI/IES LM-78-17

Illuminating Engineering Society. Approved Method: Total Luminous Flux Measurement of Lamps Using an Integrating Sphere Photometer. New York: IES; 2017.

For measurements using an integrating sphere system, the laboratory shall meet the requirements stated therein.

2.7 ANSI/IES LM-79-19

Illuminating Engineering Society. Approved Method: Optical and Electrical Measurements of Solid-State Lighting Products. New York: IES; 2019.

2.8 ANSI/IES LS-1-20

Illuminating Engineering Society. Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Apr 22).

2.9 NEMA 77-2017

National Electrical Manufacturers Association. Temporal Light Artifacts: Test Methods and Guidance for Acceptance Criteria. Arlington, Virg.: NEMA; 2017.

2.10 NEMA SSL 7A-2015

National Electrical Manufacturers Association. Phase Cut Dimming for Solid State Lighting: Basic Compatibility. Arlington, Virg.: NEMA; 2015.

3.0 Ambient and Physical Test Conditions

3.1 Ambient Temperature

The ambient temperature shall be maintained at $25\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$, and measured at a point of not more than 1 m from the device under test (DUT) and at the same mounting height as the DUT. The temperature sensor shall be shielded from direct luminous radiation from the DUT and luminous or thermal radiation from any other source. Data recorded or observations made with ambient temperatures outside of the required $25\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ are nonstandard conditions and shall be noted in the test report.

3.2 Air Movement

The incidence of air movement on the surface of a device under test (DUT) may substantially alter electrical and

photometric values. Airflow around the device under test should be such that normal convective airflow induced by the DUT is not affected.

3.3 Vibration

The natural or induced vibrations present in the laboratory should not compromise the validity of the measurement results. Vibration effects associated with the photodetector and with the device under test should be minimized in order to minimize external luminous noise that can affect test results.

3.4 Humidity

Relative humidity values greater than approximately 65% can lead to corrosion effects in some instruments, and values below approximately 10% can lead to electrostatic effects. Therefore, laboratory humidity should be monitored and kept between 10% and 65%.

3.5 Stray Light

Stray light from sources other than the DUT shall be minimized with any TLM setup.

4.0 Test Equipment

Test equipment should be selected to meet desired sampling, accuracy, detection limits, repeatability, reproducibility, and robustness. Equipment selection is always a balance between technical requirements, risks, and economic considerations.

A typical TLM measurement setup will consist of a stable power source driving an optional dimmer and a device under test (DUT; the test light source). The DUT is located in an integrating sphere or a test chamber that eliminates stray light from other possible light sources. A photocell gathers the analog luminous flux signal from the source. A transimpedance amplifier converts the photodiode current to a voltage signal. A low-pass filter is utilized to eliminate higher-frequency components that are not utilized in any TLA calculations. The analog signal is digitized and converted to a digital data file, where calculations for TLA metrics are then made.

This section identifies minimum equipment considerations for accurate and repeatable measurements associated with luminous flux waveforms utilized in calculations of temporal light artifacts.

A general test setup for measuring luminous flux waveforms from a light source consists of:

- Power supply
- Power analyzer
- Dimmer (optional)
- Test chamber
- Oscilloscope or digitizer
- Transimpedance amplifier
- Low-pass filter
- Photodetector

4.1 Power Analyzer

Electrical measurement equipment characteristics shall meet the requirements identified in the applicable IES document (see normative references).

4.2 Dimmer

For dimmable DUTs, TLM testing under dimming conditions is recommended. Testing with a standardized dimmer circuit (NEMA SSL 7A-2015; refer to **Section 2**) or with application specific dimmers (commercial dimmers) is allowed.

For non-dimmable DUTs, nominal input voltage to the DUT shall be utilized, recorded and included in the test report.

Dimming levels are expressed as a percentage of the maximum light output.

4.2.1 Forward Phase-Cut Dimming. For testing of DUTs at dimmed levels, the dimmer circuit identified in NEMA SSL 7A-2015 (refer to **Section 2**) shall be utilized.

4.2.2 Application Specific Dimming. A dimmer integral to the DUT is considered to be an application specific dimmer.

The DUT can be operated with dimmers other than the dimmer circuit identified in NEMA SSL 7A-2015 (refer to **Section 2**). The DUT can be tested with commercial dimmers and with differing dimming levels. Under these measurement conditions, the dimmer utilized, and the dimming levels tested shall be recorded and included in the test report.

4.3 Test Chamber

A test chamber that shields the DUT from light from sources other than the DUT is required. An integrating sphere, or an optically isolated chamber or dark room can be utilized.

4.3.1 Integrating Sphere. An integrating sphere utilized with a photodiode can be used as a TLM test chamber. Integrating sphere characteristics shall meet the requirements identified in IES LM-78-17.

4.3.2 Optically Isolated Chamber or Dark Room. An optically isolated chamber or dark room can be utilized as a TLM test chamber. The requirements for this chamber or dark room are described below.

4.3.2.1 DUT-to-Detector Distance. The selected distance shall provide sufficient signal at the lowest dimming level tested. If a change in distance is necessitated, the distance at each dimming condition shall be recorded. Attention shall be taken to ensure that the photocell does not saturate under any measurement condition.

For LED based DUTs where all the device LEDs utilize a common input voltage waveform with the same phase angle (e.g., those DUTs with common drivers), the distance from the source to the detector shall be such as will ensure that the photocell does not saturate at 100% source light output (the undimmed condition).

For DUTs where the drive conditions are unknown, or for DUTs containing banks of LEDs driven by more than one driver or level (e.g., AC LEDs), the distance from the DUT luminous surface to the detector shall be no less than 2.0 times the maximum luminous dimension of the DUT.

4.3.2.2 Stray Light Control. The field of view of the photodetector should be limited (e.g., using aperture screens) in order to shield from reflected stray light. Use of a light trap to block the stray light within the field of view of the photodetector is recommended.

The use of low reflectance materials (such as black felt) for the wall and floor surfaces is recommended.

4.4 Oscilloscope or Digitizer

Equipment utilized for digitizing luminous flux signals shall be capable of the following sampling frequencies:

- A minimum sampling rate for high-frequency luminous flux signals of at least 30,000 samples per second (30 kS/s); this corresponds to at least 10 times the pass band edge frequency as defined in **Section 4.6**. This is typically required for stroboscopic effect analysis. A minimum data collection duration of 1 s is required.
- A sampling minimum rate for low-frequency luminous flux signals of at least 2.5 kS/s is required. This is typically required for flicker effect analysis. When utilizing a standard power supply, a minimum data collection duration of 60 s shall be used. When utilizing a non-standard power supply, a minimum data collection duration of 180 s is required.

For frequencies between 10 and 20 Hz, flicker can be perceived for modulation depths less than 0.3% (CIE TN 006). A minimum resolution of analog-to-digital converter used shall be 12-bit. This considers cases where the peak signal is significantly lower than that of the full scale, which provides 0.025% resolution in full scale. Furthermore, the noise level of the detector and the amplifier should be low enough for a reliable measurement of such modulations.

4.5 Transimpedance Amplifier

A transimpedance amplifier is typically utilized in conjunction with the selected photocell as an analog current-to-voltage converter. The transimpedance amplifier used shall have a minimum bandwidth (IEEE 1057) for use from DC to 3 kHz. The relative gain variation in this range shall be less than 1%.

4.6 Low-Pass Filter

The pass band edge frequency for the low-pass filter shall be no less than 3 kHz. The stopband edge shall have a minimum attenuation of at least 60 dB at a frequency of 0.5 times the sampling rate frequency. Signals shall pass through the filter with passband attenuation that does not exceed ± 0.1 dB compared to 0 Hz. The use of a Butterworth low-pass filter with a 48-dB-per-octave roll-off is recommended.

4.7 Photodetector

The photodetector* shall have relative spectral responsivity well matched to the $V(\lambda)$ function and shall have an f_1' value of 5% or less. In addition, the device shall have a rise time of 10 microseconds or less, in agreement with JA10-2 (see **CEC JA10**).

The viewing angle of the photodetector shall be such that all spatial components related to TLM are captured. The photodetector shall have a cosine response f_2 of 5% or less.

4.8 Voltage Differential Probe

If utilized in the measurement process, voltage differential probes shall have a minimum bandwidth of at least 10 kHz and a common-mode rejection ratio of at least 80 dB at 50 or 60 Hz.

5.0 Measurement Process

The process described in this section shall be followed to gather the luminous flux waveform data. Calculations from the luminous flux waveform data are utilized to determine TLA metrics. **Informative Annex A** describes metrics that have been proposed or are utilized in the industry today.

* A photometer head is acceptable for utilization as a photodetector, provided it meets the criteria outlined in this document. *Photometer head* refers to a unit containing a detector, a $V(\lambda)$ -correction filter, and any additional components (e.g., aperture, diffuser, amplifier) within the unit.

5.1 Setup

The DUT shall be placed in an integrating sphere or in an optically isolated chamber or dark room, as prescribed in **Section 4.3**. The chamber or dark room shall maintain stray light from external sources at less than 0.5% of average intensity of the DUT over time. An analysis of the stray light with regard to the frequency range of interest should be performed. Any component in this frequency range may produce error and should be addressed.

For the optically isolated chamber or dark room, the detector shall be aligned to the center of the DUT and normal to the light-emitting surface of the DUT to the extent practical. Photographs or diagrams of the setup should be recorded and included in the test report.

5.2 Input Power

5.2.1 Standard Input Power. Power supplied to the DUT or to the dimmer circuit, if applicable, shall conform to the requirements identified in the applicable IES document (see **Section 2 Normative References**).

5.2.2 Nonstandard Input Power. For nonstandard power supplied to the DUT or to the dimmer circuit, the input voltage waveform, frequency, and total harmonic distortion shall be recorded and included in the test report.

5.3 Input Voltage

The DUT shall be operated at the rated voltage (AC or DC) according to the specification of the device for its normal use. For drivers with multiple input voltages, the voltage value utilized during the test shall be reported. When operated with a dimmer, the input voltage to the dimmer shall be at the rated voltage according to the specification of the dimmer for its normal use. Dimming levels shall be controlled with the dimmer control and not by adjusting the input voltage to the test setup.

The DUT can be operated with nonstandard input voltage waveforms. Under these measurement conditions, the input voltage waveform characteristics shall be recorded and included in the test report.

5.4 Dimming

It is recognized that TLM assessments can include a variety of dimming conditions. Therefore, for each TLA test that utilizes a dimmer circuit, the dimmer utilized, all dimming levels recorded as a percentage of 100% full light output,* and power input characteristics to the dimmer shall be recorded and included in the test report.

It is recommended that at a minimum, measurements should be taken at the following percentages of full light output: 100%, 20% ($\pm 2\%$), and the minimum dimming level for the dimmer control. When the minimum light output of the system is greater than 20% of full light output, then the luminous flux waveform measurements are taken at the minimum light output. All dimming levels utilized shall be recorded.

5.5 Stabilization

The DUT shall be operated for a minimum of 5 minutes, followed by an examination of stability of the DUT. The light output shall be stabilized in advance of taking measurements at each dimming level. Light output shall be considered stabilized when 4 consecutive measurements taken at one-minute intervals deviate by no more than 0.5%. If the DUT does not reach the 0.5% stability after 20 minutes of operation, the luminous flux waveform measurement, the time, and the stability value for the last 4 consecutive minutes shall be recorded.

* 100% full light output is defined as operating the light source at the maximum setting provided by the dimmer control.

5.6 Measurement

Once stabilization has been reached, the luminous flux waveform(s) shall be sampled for the duration of the measurement(s) as outlined in **Section 4.4**.

The luminous flux waveform data shall contain a minimum of 4 significant digits. It is recommended that the recorded data utilize the number of significant digits in the output from the measurement equipment.

Time stamped data shall be stored in a comma separated value (CSV) format, XML, or equivalent.

5.6.1 Dark signal. *Dark signal* is defined as any background light not from the DUT, along with any electronic instrumentation-generated offset or noise.

A dark signal shall be measured and subtracted from the measurement data collected in the process described in **Section 5.5**. The process for measuring the dark signal is as follows:

- After data from the measurement described in **Section 5.6** are collected, the DUT is switched off. After a sufficiently long time that no light is emitting from the DUT (typically on the order of 10 s), the dark signal is recorded.
- If the dark signal or average luminous flux waveform value is greater than 15%, then the measurement shall be rejected.
- If the dark signal or average luminous flux waveform value is less than or equal to 15%, then the dark signal shall be subtracted from the luminous flux waveform data recorded in the process described in **Section 5.5**. The resulting time-stamped luminous flux waveform data set shall be stored in a comma separated value (CSV) format, XML, or equivalent.

6.0 Luminous Flux Waveform Data File

It is recommended that the luminous flux waveform data be formatted in accordance with W3C Extensible Markup Language (XML) 1.1 Recommendation and the W3C XML Schema Definition Language (XSD). This allows end users to utilize and view the data directly without the need for proprietary software.

In the event of any ambiguity or discrepancy in this document with respect to the textual specification and the XML Schema, the XML Schema shall take precedence.

In the event of any ambiguity or discrepancy in this document with respect to the description of XML or XML Schema, the W3C Recommendations shall take precedence.

The Luminous Flux Waveform Data document is defined in terms of the W3C XML Schema presented in **Annex B**.

The XML document is separated into four elements:

- The required `Version` element, which identifies the XML schema version
- The required `Header` element, which contains all generic information about the light source
- The required `Equipment` element, which contains information about the test chamber, measurement, and auxiliary equipment used to perform the TLM measurements

- The required `TLMDATA` element, which contains information about the TLM measurements, including generic information, intensity distribution data, and spectral power distribution data

XML Schema can be found in **Informative Annex B**, and an associated sample TLM XML can be found in **Informative Annex C** of this document.

6.1 Version Element

The required `Version` element identifies the XML schema version. It is required to be "1.1".

6.2 Header Element

The required `Header` element is the parent of the document header section. This section does not include TLM data. Header elements are listed in **Table 6-1**.

Table 6-1. Header Element Fields

Element Description	Element Name	Data Type	Required	Document Section
Manufacturer	Manufacturer	xs:string	Yes	6.2.1
Product Number	ProductNumber	xs:string	Optional	6.2.2
Description	Description	xs:string	Yes	6.2.3
Configuration	Configuration	xs:string	Yes	6.2.4
Laboratory	Laboratory	xs:string	Yes	6.2.5
Report Number	ReportNumber	xs:string	Yes	6.2.6
Report Date	ReportDate	xs:date	Yes	6.2.7
Comment	Comment	xs:string	Optional	6.2.8

6.2.1 Manufacturer Element. The required `Manufacturer` element identifies the manufacturer of the DUT or light source.

6.2.2 Product Number Element. The optional `ProductNumber` element identifies the manufacturer's product or model number.

6.2.3 Description Element. The required `Description` element contains a text description of the DUT or light source.

6.2.4 Configuration. The required `Configuration` element contains a text description of the Driver(s), driver settings, ballast, or other relevant information.

6.2.5 Laboratory Element. The required `Laboratory` element identifies the name of the testing laboratory that performed the data measurements. If the data were generated from a computer simulation and not from a laboratory test, this field shall contain the name of the company that generated the data.

6.2.6 Report Number Element. The required `ReportNumber` element identifies the testing laboratory report number.

6.2.7 Report Date Element. The required `ReportDate` element identifies the testing laboratory report date using the XSD Date data type, YYYY-MM-DD.

6.2.8 Comment Element. The optional `Comment` element provides additional information relating to the tested and reported data.

Multiple `Comment` elements are allowed.

6.3 Equipment Element

The required `Equipment` element is the parent of the information on the test chamber, measurement, and auxiliary equipment, and the setup media. Elements are detailed in **Table 6-2**.

Table 6-2. Equipment Element Fields

Element Description	Element Name	Data Type	Required	Document Section
Test Chamber	TestChamber	xs:string	Yes	6.3.1
Measurement Equipment	MeasurementEquipment	XML Element	Yes	6.3.2
Auxiliary Equipment	AuxiliaryEquipment	XML Element	Optional	6.3.3
Setup Media	SetupMedia	XML Element	Optional	6.3.4

6.3.1 Test Chamber Element. The required `TestChamber` element contains a description of the test chamber used, including distances, for the TLM measurement data.

6.3.2 Measurement Equipment Element. The required `MeasurementEquipment` element is the parent of the information on the measurement equipment used for the TLM measurement data. This element contains information that is specific to the measurement equipment data. Elements are detailed in **Table 6-3**.

Multiple `MeasurementEquipment` elements are allowed.

Table 6-3. Measurement Equipment Fields

Element Description	Element Name	Data Type	Required	Document Section
Description	Description	xs:string	Yes	6.3.2.1
Identification	Identification	xs:string	Yes	6.3.2.2

6.3.2.1 Description Element. The required `Description` element describes the equipment used for the TLM measurement.

6.3.2.2 Identification Element. The required `Identification` element identifies the equipment used for the TLM measurement (e.g., serial number, asset number).

6.3.3 Auxiliary Equipment Element. The optional `AuxiliaryEquipment` element is the parent of the information on the auxiliary equipment or external controls used, if any, for the TLM measurement data. This element contains information that is specific to the auxiliary equipment or external controls data. The `AuxiliaryEquipment` element includes dimmer make and model information, or whether the dimmer is the NEMA SSL 7A reference dimmer (refer to **Section 2**). Elements are detailed in **Table 6-4**.

Multiple `AuxiliaryEquipment` elements are allowed.

Table 6-4. Auxiliary Equipment Fields

Element Description	Element Name	Data Type	Required	Document Section
Description	Description	xs:string	Yes	6.3.3.1
Identification	Identification	xs:string	Yes	6.3.3.2

6.3.3.1 Description Element. The required `Description` element describes the equipment used for the TLM measurement.

6.3.3.2 Identification Element. The required `Identification` element identifies the equipment used for the TLM measurement (e.g., serial number, asset number).

6.3.4 Setup Media Element. The optional `SetupMedia` element is the parent of the information on any media (e.g., pictures, diagrams) used to show the setup for the TLM measurement. Elements are detailed in **Table 6-5**.

Multiple `SetupMedia` elements are allowed.

Table 6-5. Auxiliary Equipment Fields

Element Description	Element Name	Data Type	Required	Document Section
Description	Description	xs:string	Yes	6.3.4.1
File Name	FileName	xs:string	Yes	6.3.4.2

6.3.4.1 Description Element. The required `Description` element describes the media showing the setup for the TLM measurement.

6.3.4.2 File Name Element. The required `FileName` element identifies the file name, including its file extension, which contains the media showing the setup for the TLM measurement.

6.4 TLM Data Element

The required `TLMData` element is the parent of the TLM data. Elements are detailed in **Table 6-6**.

Table 6-6. TLM Data Element Fields

Element Description	Element Name	Data Type	Required	Document Section
Nonstandard Conditions	NonstandardConditions	xs:string	Optional	6.4.1
Additional Information	AdditionalInformation	xs:string	Optional	6.4.2
Uncertainty Statement	UncertaintyStatement	xs:string	Optional	6.4.3
Rated Electrical Input	RatedElectricalInput	xs:string	Yes	6.4.4
Electrical Frequency	ElectricalFrequency	xs:int	Optional	6.4.5
DUT Measurement Data	DUTMeasurementData	XML Element	Yes	6.4.6

6.4.1 Nonstandard Conditions Element. The optional `NonstandardConditions` element describes any nonstandard conditions for the TLM measurement.

6.4.2 Additional Information Element. The optional `AdditionalInformation` element describes any additional information for the TLM measurement.

6.4.3 Uncertainty Statement Element. The optional `UncertaintyStatement` element specifies the uncertainty statement for the TLM measurement.

6.4.4 Rated Electrical Input Element. The required `RatedElectricalInput` element specifies the rated electrical input (e.g., 120 V AC) for the TLM measurement.

6.4.5 Electrical Frequency Element. The optional `ElectricalFrequency` element specifies the frequency value for the TLM measurement. This is only applicable when the Rated Electrical Input element is set to "AC".

6.4.6 DUT Measurement Data Element. The required `DUTMeasurementData` element is the parent of the DUT TLM measurement data. Elements are detailed in **Table 6-7**.

Multiple `DUTMeasurementData` elements are allowed.

Table 6-7. DUT Measurement Data Element Fields

Element Description	Element Name	Data Type	Required	Document Section
DUT Identification	DUTIdentification	xs:string	Yes	6.4.6.1
Test Operator	TestOperator	xs:string	Yes	6.4.6.2
Test Burn Position	TestBurnPosition	xs:string	Yes	6.4.6.3
TLM Data	TLMDData	XML Element	Yes	6.4.6.4

6.4.6.1 DUT Identification Element. The required `DUTIdentification` element specifies the identification of the DUT.

6.4.6.2 Test Operator Element. The required `TestOperator` element specifies the test operator for the TLM measurement.

6.4.6.3 Test Burn Position Element. The required `TestBurnPosition` element specifies the test burn position for the TLM measurement.

6.4.6.4 TLM Data Element. The required `TLMDData` element is the parent of the DUT TLM measurement data and calculation results. Elements are detailed in **Table 6-8**.

Multiple `TLMDData` elements are allowed.

Table 6-8. TLM Data Element Fields

Element Description	Element Name	Data Type	Required	Document Section
Test Date	TestDate	xs:date	Yes	6.4.6.4.1
Dark Signal	DarkSignal	xs:decimal	Yes	6.4.6.4.2
Ambient Temperature	AmbientTemperature	xs:decimal	Yes	6.4.6.4.3
Ambient Relative Humidity	AmbientHumidity	xs:decimal	Optional	6.4.6.4.4
Operating Time Prior Test	OperatingTimePriorTest	xs:int	Yes	6.4.6.4.5
Test Operating Time	TestOperatingTime	xs:int	Yes	6.4.6.4.6
Stabilization Time	StabilizationTime	xs:int	Yes	6.4.6.4.7
Dimming Level Percentage	DimmingLevelPercentage	xs:int	Yes	6.4.6.4.8
Electrical Input Voltage	ElectricalInputVoltage	xs:decimal	Yes	6.4.6.4.9
Electrical Input Current	ElectricalInputCurrent	xs:decimal	Yes	6.4.6.4.10
Electrical Input Power	ElectricalInputPower	xs:decimal	Yes	6.4.6.4.11
Flicker Index	FlickerIndex	xs:decimal	Optional	6.4.6.4.12
Jitter	Jitter	xs:decimal	Optional	6.4.6.4.13
Maximum Light Output Waveform	MaximumLightOutputWaveform	xs:decimal	Optional	6.4.6.4.14
Modulation Depth	ModulationDepth	xs:decimal	Optional	6.4.6.4.15
Unfiltered Percent Flicker	UnfilteredPercentFlicker	xs:decimal	Optional	6.4.6.4.16
40-Hz Filtered Percent Flicker	40HzFilteredPercentFlicker	xs:decimal	Optional	6.4.6.4.17
90-Hz Filtered Percent Flicker	90HzFilteredPercentFlicker	xs:decimal	Optional	6.4.6.4.18
200-Hz Filtered Percent Flicker	200HzFilteredPercentFlicker	xs:decimal	Optional	6.4.6.4.19
400-Hz Filtered Percent Flicker	400HzFilteredPercentFlicker	xs:decimal	Optional	6.4.6.4.20
1,000-Hz Filtered Percent Flicker	1000HzFilteredPercentFlicker	xs:decimal	Optional	6.4.6.4.21
Phantom Array Effect	PhantomArrayEffect	xs:decimal	Optional	6.4.6.4.22
Short-Term Flicker Indicator (Pst)	Pst	xs:decimal	Optional	6.4.6.4.23
Stroboscopic Effect	StroboscopicEffect	xs:decimal	Optional	6.4.6.4.24
Stroboscopic Visibility Measurement (SVM)	SVM	xs:decimal	Optional	6.4.6.4.25
Temporal Light Artifact (TLA)	TLA	xs:decimal	Optional	6.4.6.4.26
Flicker Perception Metric (MP)	MP	xs:decimal	Optional	6.4.6.4.27
Luminous Flux Signal Data	LuminousFluxSignalData	XML Element	Required	6.4.6.4.28
Other Data	OtherData	XML Element	Optional	6.4.6.4.29

6.4.6.4.1 Test Date Element. The required `TestDate` element specifies the test date for the TLM measurement using the XSD Date data type, YYYY-MM-DD.

6.4.6.4.2 Dark Signal Element. The required `DarkSignal` element specifies the dark signal for the TLM measurement (typically counts).

6.4.6.4.3 Ambient Temperature Element. The required `AmbientTemperature` element specifies the ambient temperature for the TLM measurement.

6.4.6.4.4 Ambient Humidity Element. The optional `AmbientHumidity` element specifies the ambient relative humidity for the TLM measurement.

6.4.6.4.5 Operating Time Prior Test Element. The required `OperatingTimePriorTest` element specifies the operating time prior to the TLM measurement, in minutes.

6.4.6.4.6 Test Operating Time Element. The required `TestOperatingTime` element specifies the operating time of the TLM measurement, in minutes.

6.4.6.4.7 Stabilization Time Element. The required `StabilizationTime` element specifies the stabilization time for the TLM measurement, in minutes.

6.4.6.4.8 Dimming Level Percentage Element. The required `DimmingLevelPercentage` element specifies the dimming level for the TLM measurement as a percentage of the total light output.

6.4.6.4.9 Electrical Input Voltage Element. The required `ElectricalInputVoltage` element specifies the electrical input voltage to the DUT during the TLM measurement.

6.4.6.4.10 Electrical Input Current Element. The required `ElectricalInputCurrent` element specifies the electrical input current to the DUT during the TLM measurement.

6.4.6.4.11 Electrical Input Power Element. The required `ElectricalInputPower` element specifies the electrical input power to the DUT during the TLM measurement.

6.4.6.4.12 Flicker Index Element. The optional `FlickerIndex` element specifies the value of the unfiltered flicker index for the TLM measurement.

6.4.6.4.13 Jitter Element. The optional `Jitter` element specifies the jitter value for the TLM measurement.

6.4.6.4.14 Maximum Light Output Waveform Element. The optional `MaximumLightOutputWaveform` element specifies the maximum light output waveform value for the TLM measurement.

6.4.6.4.15 Modulation Depth Element. The optional `ModulationDepth` element specifies the modulation depth value for the TLM measurement.

6.4.6.4.16 UnfilteredPercentFlicker Element. The optional `PercentFlicker` element specifies the percent flicker value for the TLM measurement.

6.4.6.4.17 40HzFilteredPercentFlicker Element. The optional `40HzFilteredPercentFlicker` element specifies the low-pass filtered percent flicker value for the TLA measurement with a 40-Hz cutoff frequency.

6.4.6.4.18 90HzFilteredPercentFlicker Element. The optional `90HzFilteredPercentFlicker` element specifies the low-pass filtered percent flicker value for the TLM measurement with a 90-Hz cutoff frequency.

6.4.6.4.19 200HzFilteredPercentFlicker Element. The optional `200HzFilteredPercentFlicker` element specifies the low-pass filtered percent flicker value for the TLM measurement with a 200-Hz cutoff frequency.

6.4.6.4.20 400HzFilteredPercentFlicker Element. The optional `400HzFilteredPercentFlicker` element specifies the low-pass filtered percent flicker value for the TLM measurement with a 400-Hz cutoff frequency.

6.4.6.4.21 1000HzFilteredPercentFlicker Element. The optional `1000HzFilteredPercentFlicker` element specifies the low-pass filtered percent flicker value for the TLM measurement with a 1,000-Hz cutoff frequency.

6.4.6.4.22 Phantom Array Effect Element. The optional `PhantomArrayEffect` element specifies the value of the phantom array effect for the TLM measurement.

6.4.6.4.23 Short Term Flicker Indicator (Pst) Element. The optional `Pst` element specifies the value of the short-term flicker indicator for the TLM measurement.

6.4.6.4.24 Stroboscopic Effect Element. The optional `StroboscopicEffect` element specifies the value of the stroboscopic effect for the TLM measurement.

6.4.6.4.25 Stroboscopic Visibility Measurement (SVM) Element. The optional `SVM` element specifies the value of the stroboscopic visibility measurement for the TLM measurement.

6.4.6.4.26 Temporal Light Artifact (TLA) Element. The optional `TLA` element specifies the temporal light artifact value for the TLA measurement.

6.4.6.4.27 Flicker Perception Metric (MP) Element. The optional `MP` element specifies the flicker perception metric value for the TLM measurement.

6.4.6.4.28 Luminous Flux Signal Data Element. The required `LuminousFluxSignalData` element is the parent of the DUT TLM measurement luminous flux signal data. Elements are detailed in **Table 6-9**.

Table 6-9. Luminous Flux Signal Data Element Fields

Element Description	Element Name	Data Type	Required	Document Section
Sampling Interval	<code>SamplingInterval</code>	xml:decimal	Required	6.4.6.4.28.1
Luminous Flux Signal	<code>LuminousFluxSignal</code>	xml:decimal	Required	6.4.6.4.28.2

6.4.6.4.28.1 Sampling Interval Element. The required `SamplingInterval` element specifies the value of the sampling interval for the TLM measurement luminous flux signal data.

6.4.6.4.28.2 Luminous Flux Signal Element. The required `LuminousFluxSignal` element specifies the value of the luminous flux signal for the TLM measurement luminous flux signal data.

Multiple `LuminousFluxSignal` elements are allowed.

6.4.6.4.29 Other Data Element. The optional `OtherData` element is the parent of any other data relevant to the DUT TLM measurement. Elements are detailed in **Table 6-10**.

Multiple `OtherData` elements are allowed.

Table 6-10. Other Data Element Fields

Element Description	Element Name	Data Type	Required	Document Section
Description	Description	xml:string	Required	6.4.6.4.29.1
Value	Value	xml:decimal	Required	6.4.6.4.29.2

6.4.6.4.29.1 Description Element. The required `Description` element specifies the description of the other data relevant to the TLA measurement.

6.4.6.4.29.2 Value Element. The required `Value` element specifies the value of the other data relevant to the TLA measurement.

7.0 Uncertainty Analysis

An analysis of components that affect the uncertainty of the luminous flux waveform measurements shall be performed, with an estimate of the magnitude of each component identified. The basic measurement equation associated with the luminous flux waveform (**Equation 7-1**) can be constructed from the treatment shown in **Figure 7-1**:

$$u_D(t) = (R_V(f) \cdot E_V(t) + \Delta_i) \cdot A(f) \cdot G(f) + \Delta_{digit} \quad (7-1)$$

Uncertainty components that should be examined, as appropriate, include:

- Photometer – non-linearity (Δ_i)
- Illuminance responsivity of the photometer [$R_V(f)$]
- Current to voltage converter (CVC) amplifier – gain flatness [$\Delta A(f)$]
- The low pass filter – flatness [$\Delta G(f)$]
- Analogue to digital converter – resolution and timing jitter (Δ_{digit})
- Dark signal [$u_{D,dark}(t)$]

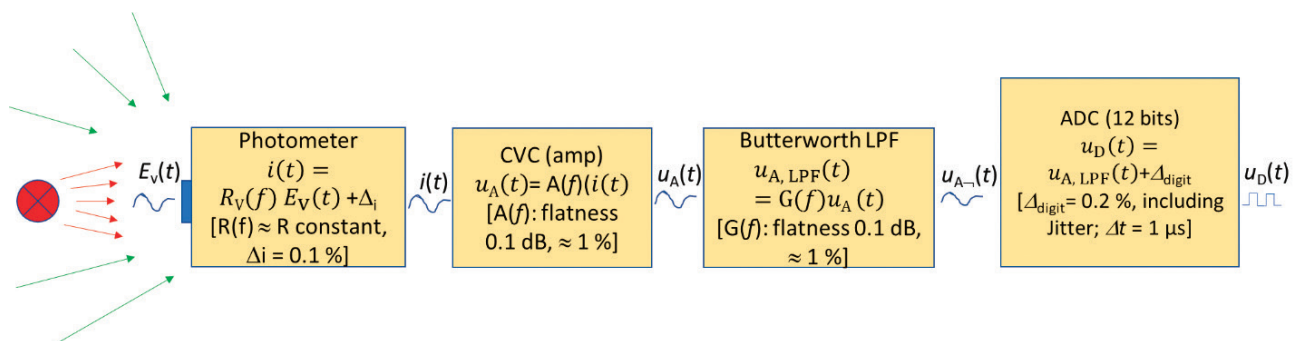


Figure 7-1. Methodology for constructing the luminous flux waveform equation.

In addition, the luminous flux waveform [$E_V(t)$] may contain nonlinearities as a function of analogue or digital dimming levels.

An example uncertainty analysis can be found in **Informative Annex D**.

Informative Annex A – Descriptions of TLA Metrics

Filtered Percent Flicker (Percent Amplitude Modulation)

The filtered percent flicker metric is a calculation of percent flicker on a luminous flux data set that has high frequency data filtered out. Fourier analysis transforms time-series luminous flux data into the frequency domain and frequency data above a predetermined cut-off frequency is filtered out. An inverse Fourier transformation places the filtered frequency data back into the time domain, yielding filtered time domain data. Filtered percent flicker is calculated on the resulting filtered time domain data over the full sampling duration.

Common filter frequency evaluation is performed for five different cut-off frequencies: frequencies under 40 Hz (for which data above 40 Hz is set to 0), and frequencies under 90 Hz, 200 Hz, 400 Hz, and 1,000 Hz. Data is transferred back into the time domain using an inverse Fourier transform, and percent amplitude modulation is then calculated. [Adapted from CEC JA10].

Flicker

Per CIE TN 006:2016:

[The] perception of visual unsteadiness induced by a light stimulus, the luminance or spectral distribution of which fluctuates with time, for a static observer in a static environment

Note: The fluctuations of the light stimulus with time include periodic and non-periodic fluctuations and may be induced by the light source itself, the power source, or other influencing factors. Flicker is perceived up to frequencies of approximately 80 Hz, depending on the conditions.

Flicker Index

Per ANSI/IES LS-1-20:

A measure of the cyclic variation in output of a light source taking into account the waveform of the light output. It is the ratio of the area under the light output curve that is above the average light output level to the total area under the light output curve for a single cycle.

Flicker Perception Metric (MP)

Per ASSIST recommends Volume 11, Issue 3:

The ASSIST Flicker Perception Metric is an objective method to assess the perception of flicker directly observed from light sources. Calculation of the metric starts with a relative light output waveform measurement and finishes with a single numerical result indicating whether the amount of flicker is above or below the human threshold of perception. The result can also be expressed as the probability of an observer detecting flicker from the source. The metric is applicable to any waveform shape and frequency.

Note: Effective October 1, 2017, the U.S. Environmental Protection Agency (EPA) is requiring that relative light output waveform measurement(s) be taken for all lamps seeking ENERGY STAR certification and that the value of MP be reported.

Jitter

Per NEMA 77-2017:

Observed in phase-cut dimmable systems, the difference between minimum and maximum times between successive leading phase-cut edges of the same polarity, measured over hundreds of cycles. *Positive jitter* is the variation in times between positive leading edges; *negative jitter* is the variation in times between negative leading edges. Jitter should typically be observed over many hundreds of cycles, and the highest observed jitter value is used. Jitter does not need to be evenly distributed and centered on the firing angle setting. For example, 999 cycles where the firing edge is perfectly reproduced followed by a single edge that is 100 µs late = 100 µs jitter.

Maximum Light Output Waveform (MLOW)

Per NEMA 77-2017:

The light output waveform produced by a light source when connected directly to nominal mains voltage without a dimmer.

Modulation Depth

Per CIE TN 006:2016 (see **Section 2**):

Within a period of time, a waveform reaches a maximum intensity (y_{max}) and a minimum intensity (y_{min}) and has a time average ($\bar{y}$). Modulation depth (D_M) is defined as:

$$D_M = \frac{y_{max} - y_{min}}{y_{max} + y_{min}}$$

For waveforms for which the time-average ($\bar{y}$) is equal to $(y_{max} + y_{min})/2$, the modulation depth is equal to:

$$D_M = (y_{max} - y_{min})/2\bar{y}$$

Modulation depth is often expressed as a percentage.

Percent Flicker

Per ANSI/IES LS-1-20:

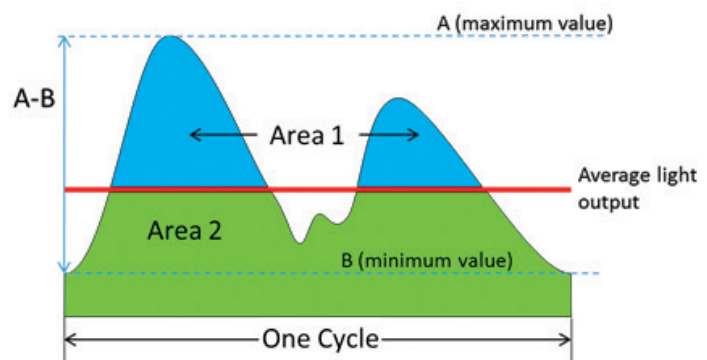
A relative measure of the cyclic variation in output of a light source (percent modulation). It is given by the expression $100 \times (A - B)/(A + B)$, where A is the maximum and B is the minimum output during a single cycle (see **Figure A-1**). This metric is also known as *amplitude modulation unfiltered*.

Phantom Array Effect (ghosting)

Per CIE TN 006:2016:

[The] change in perceived shape or spatial positions of objects, induced by a light stimulus the luminance or spectral distribution of which fluctuates with time, for a non-static observer in a static environment

Note: The phantom array effect is a type of temporal light artefact.



$$\text{Percent Flicker} = 100 \times \frac{A - B}{A + B}$$

$$\text{Flicker Index} = \frac{\text{Area 1}}{\text{Area 1} + \text{Area 2}}$$

Figure A-1. Formulas for percent flicker (also called modulation depth, or modulation percent) and flicker index.

Example: When making a saccade over a small light source having a square periodic luminance fluctuation, the light source is perceived as a series of spatially extended light spots.

Short-Term Flicker Indicator (Pst)

Per IEC TR 61547; IEC 61000-3-3; IEC 61000-4-15:

A measure of flicker evaluated over a specified time interval of a relatively short duration. Note: The duration is typically 10 minutes, in accordance with Reference [IEC 61000-3-3]. Reference [IEC TR 61547] recommends a minimum test duration of 3 minutes.

Note: Effective October 1, 2017, the U.S. Environmental Protection Agency (EPA) is requiring that relative light output waveform measurement(s) be taken for all lamps seeking ENERGY STAR certification and that the value of Pst be reported.

Stroboscopic Effect

Per CIE TN 006:2016:

[Stroboscopic effect is the] change in motion perception induced by a light stimulus the luminance or spectral distribution of which fluctuates with time, for a static observer in a non-static environment

Note 1: The stroboscopic effect is a type of temporal light artifact.

Example 1: For a square periodic luminance fluctuation, moving objects are perceived to move discretely rather than continuously.

Example 2: If the frequency of a periodic luminance fluctuation coincides with the frequency of a rotating object, the rotating object is perceived as static.

Stroboscopic Visibility Measure (SVM)

Per NEMA 77-2017 and CIE TN 006:2016:

A metric, calculated from a light waveform, that indicates the threshold for visibility of the stroboscopic effect.

The relevant frequencies for stroboscopic effects range from 80 Hz to approximately 2,000 Hz. SVM is a method for quantifying the visibility of stroboscopic effects. The method is based on experimental work published by Perz et al. A sensitivity curve was derived, and a weighted Fourier method is used to determine the metric.

$$SVM = \sqrt[3.7]{\sum_{m=1}^{\infty} \left(\frac{C_m}{T_m} \right)^{3.7}}$$

C_m is the m -th Fourier component of the light output waveform. T_m is the visibility threshold at the m -th frequency, which weights the Fourier components. $T_m(f)$ is described by:

$$T_m(f) = \left(\frac{1}{1 + e^{-a(f-b)}} + 20e^{-f/10} \right), \text{ expressed as a percentage,}$$

where:

f = frequency in Hz
 a = 0.00518 s
 b = 306.6 Hz

Note: Effective October 1, 2017, the U.S. Environmental Protection Agency (EPA) is requiring that relative light output waveform measurement(s) be taken for all lamps seeking ENERGY STAR certification and that the value of SVM be reported.

Temporal Light Artifact (TLA)

Per CIE TN 006:2016:

The change in visual perception, induced by a light stimulus the luminance or spectral distribution of which fluctuates with time, for a human observer in a specified environment

Note: The change of visual perception is a result of comparing the visual perception of the environment lit by the modulated light to the visual perception of the same person in the same environment, when the environment is lit by non-modulated light.

Informative Annex B – XML Schema

Note: An example of an XML Schema can be found in **Informative Annex C**.

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">

  <xs:complexType name="EquipmentType" >
    <xs:sequence>
      <xs:element name="Description" type="xs:string" />
      <xs:element name="Identification" type="xs:string" />
    </xs:sequence>
  </xs:complexType>

  <xs:element name="IESLMXX">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="Version" type="xs:string" fixed="1.1"/>
        <xs:element name="Header">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="Manufacturer" type="xs:string" />
            </xs:sequence>
          </xs:complexType>
        </xs:element>
        <xs:element name="ProductNumber" type="xs:string" minOccurs="0"/>
        <xs:element name="Description" type="xs:string" />
        <xs:element name="Configuration" type="xs:string" />
        <xs:element name="Laboratory" type="xs:string" />
        <xs:element name="ReportNumber" type="xs:string" />
        <xs:element name="ReportDate" type="xs:date" />
        <xs:element name="Comment" type="xs:string" minOccurs="0" maxOccurs="unbounded" />
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

```

        </xs:sequence>
    </xs:complexType>
    </xs:element>
    <xs:element name="Equipment">
        <xs:complexType>
            <xs:sequence>
<xs:element name="TestChamber" type="xs:string" />
<xs:element name="MeasurementEquipment" type="EquipmentType"
maxOccurs="unbounded" />
<xs:element name="AuxiliaryEquipment" type="EquipmentType" minOccurs="0"
maxOccurs="unbounded" />
<xs:element name="SetupMedia" minOccurs="0" maxOccurs="unbounded" >
    <xs:complexType>
        <xs:sequence>
            <xs:element name="Description" type="xs:string" />
<xs:element name="FileName" type="xs:string" />
        </xs:sequence>
    </xs:complexType>
</xs:element>
    </xs:sequence>
</xs:complexType>
    </xs:element>
    <xs:element name="TLMDData">
        <xs:complexType>
            <xs:sequence>
<xs:element name="NonStandardConditions" type="xs:string" minOccurs="0" />
<xs:element name="AdditionalInformation" type="xs:string" minOccurs="0" />
<xs:element name="UncertaintyStatement" type="xs:string" minOccurs="0" />
<xs:element name="RatedElectricalInput" type="xs:string"/>
<xs:element name="ElectricalFrequency" type="xs:string" minOccurs="0" />
<xs:element name="DUTMeasurementData" maxOccurs="unbounded" >
    <xs:complexType>
        <xs:sequence>
            <xs:element name="DUTIdentification" type="xs:string" />
            <xs:element name="TestOperator" type="xs:string" />
            <xs:element name="TestBurnPosition" type="xs:string" />
            <xs:element name="TLMDData" maxOccurs="unbounded" >
                <xs:complexType>
                    <xs:sequence>
                        <xs:element name="TestDate" type="xs:date" />
                        <xs:element name="DarkSignal" type="xs:decimal" />
                        <xs:element name="AmbientTemperature" type="xs:decimal" />
                        <xs:element name="AmbientHumidity" type="xs:decimal" minOccurs="0" />
                        <xs:element name="OperatingTimePriorTest" type="xs:int" />
                        <xs:element name="TestOperatingTime" type="xs:int" />
                        <xs:element name="StabilizationTime" type="xs:int" />
                        <xs:element name="DimmingLevelPercentage" type="xs:int" />
                        <xs:element name="ElectricalInputVoltage" type="xs:decimal" />
                        <xs:element name="ElectricalInputCurrent" type="xs:decimal" />
                        <xs:element name="ElectricalInputPower" type="xs:decimal" />
                        <xs:element name="Flicker" type="xs:decimal" minOccurs="0"/>
                        <xs:element name="FlickerIndex" type="xs:decimal" minOccurs="0"/>
                        <xs:element name="Jitter" type="xs:decimal" minOccurs="0"/>
                        <xs:element name="MaximumLightOutputWaveform" type="xs:decimal"
minOccurs="0"/>

```

```

        <xs:element name="ModulationDepth" type="xs:decimal" minOccurs="0"/>
        <xs:element name="UnfilteredPercentFlicker" type="xs:decimal"
minOccurs="0"/>
        <xs:element name="40HzFilteredPercentFlicker" type="xs:decimal"
minOccurs="0"/>
        <xs:element name="90HzFilteredPercentFlicker" type="xs:decimal"
minOccurs="0"/>
        <xs:element name="200HzFilteredPercentFlicker" type="xs:decimal"
minOccurs="0"/>
        <xs:element name="400HzFilteredPercentFlicker" type="xs:decimal"
minOccurs="0"/>
        <xs:element name="1000HzFilteredPercentFlicker" type="xs:decimal"
minOccurs="0"/>
        <xs:element name="PhantomArrayEffect" type="xs:decimal"
minOccurs="0"/>
        <xs:element name="PST" type="xs:decimal" minOccurs="0"/>
        <xs:element name="StroboscopicEffect" type="xs:decimal"
minOccurs="0"/>
        <xs:element name="SVM" type="xs:decimal" minOccurs="0"/>
        <xs:element name="TLA" type="xs:decimal" minOccurs="0"/>
        <xs:element name="MP" type="xs:decimal" minOccurs="0"/>
        <xs:element name="LuminousFluxSignalData" >
            <xs:complexType>
                <xs:sequence>
<xs:element name="SamplingInterval" type="xs:decimal" />
<xs:element name="LuminousFluxSignal" type="xs:decimal" maxOccurs="unbounded" />
                </xs:sequence>
            </xs:complexType>
        </xs:element>
        <xs:element name="OtherData" minOccurs="0"
maxOccurs="unbounded" >
            <xs:complexType>
                <xs:sequence>
<xs:element name="Description" type="xs:string" />
<xs:element name="Value" type="xs:decimal" />
                </xs:sequence>
            </xs:complexType>
        </xs:element>
    </xs:sequence>
</xs:complexType>
</xs:element>
</xs:schema>

```

Informative Annex C – Sample TLM XML Document

```
<?xml version="1.0"?>

<IESLMXX>

<Version>1.1</Version>

<Header>

<Manufacturer>Academy Lighting</Manufacturer>

<ProductNumber>XET 55529</ProductNumber>

<Description>LED 2' x 4' Troffer</Description>

<Configuration>Driver ABC, setting 123</Configuration>

<Laboratory>Apex Analytics</Laboratory>

<ReportNumber>APEX-091101-004</ReportNumber>

<ReportDate>2009-11-01</ReportDate>

<Comment>Apex Analytics</Comment>

<Comment>Apex Analytics 2</Comment>

</Header>

<Equipment>

<TestChamber>Integrating Sphere</TestChamber>

<MeasurementEquipment>

<Description>Power Meter</Description>

<Identification>S1</Identification>

</MeasurementEquipment>

<MeasurementEquipment>

<Description>Power Meter 2</Description>

<Identification>S2</Identification>

</MeasurementEquipment>

</Equipment>

<TLMDData>

<RatedElectricalInput>AC</RatedElectricalInput>
```


Approved Method: Measuring Luminous Flux Waveforms for Use in Temporal Light Artifact (TLA) Calculations

```
<ElectricalFrequency>60</ElectricalFrequency>
<DUTMeasurementData>
<DUTIdentification>T20171456/1</DUTIdentification>
<TestOperator>Mr. Smith</TestOperator>
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Informative Annex D – Uncertainty Example

The following is an example of creating an uncertainty budget for the luminous flux signal generated from a general measurement equation for reasonable system components. The uncertainty values in **Figure D-1** are conservative and have an expanded coverage factor ($k=2$). The relative uncertainty percentages in the **Table D-1** are standard (coverage factor, $k=1$). The block diagram contains expanded uncertainties with a coverage factor of $k=2$.

- Measurement equation:

$$u_D(t) = (R_V(f) \cdot E_V(t) + \Delta_i) \cdot A(f) \cdot G(f) + \Delta_{digit}$$

- Offset errors are not included which can be removed by a dark subtraction:

$$u_{meas} = u_{D,sig}(t) - u_{D,dark}(t)$$

Table D-1. Relative Standard Uncertainty of Each Component

Uncertainty Component	Relative Standard Uncertainty (%)
Photometer nonlinearity, Δ_i	0.05
CVC gain flatness, $\Delta A(f)$	0.50
LPF attenuation flatness, $\Delta G(f)$	0.50
ADC digitizing error (resolution and timing jitter), Δ_{digit}	0.10
Combined uncertainty ($k=1$) (%)	0.72
Expanded uncertainty ($k=2$) (%)	1.4

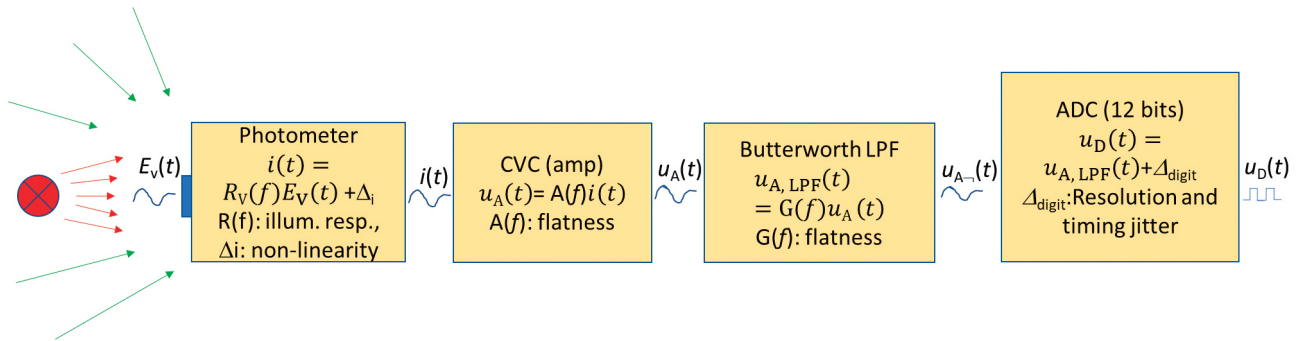


Figure D-1. Methodology for creating the luminous flux signal.

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
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Telephone: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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